Here's a **comprehensive table of all major Linux OS flavors (distributions)** used in real-time IT companies, including differences in **commands**, **package managers**, and their **typical real-world usage** in production or development environments:

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**Linux OS Flavors Comparison Table**

| Linux Flavor | Base | Package Manager | Common Commands Difference | Real-Time Use in IT Companies | Popular For |
|---|---|---|---|---|---|
| **Linux Mint** | Ubuntu | apt, dpkg | Same as Ubuntu | Developer machines, desktops | Ubuntu with better UI for desktops |
| **Zorin OS** | Ubuntu | apt | Same as Ubuntu | Desktop Linux for new users | Beginner-friendly UI |
| **Pop!_OS** | Ubuntu | apt, flatpak | apt install, flatpak run | Developer workstations | System76 devices, GPU usage |
| **openSUSE** | SUSE | zypper, rpm | zypper install, yast | Server setups in Europe, SAP environments | SAP systems, system admins |
| **SUSE Linux Enterprise (SLES)** | SUSE | zypper, rpm | zypper install, yast | Enterprise workloads, SAP, secure systems | Certified enterprise OS |
| **CentOS *(now Rocky/Alma)*** | RHEL | yum, dnf | yum install, systemctl | Legacy enterprise apps, older servers | Legacy RHEL workloads |
| **Rocky Linux** | RHEL | dnf, yum | dnf install, firewall-cmd | Modern RHEL alternative in enterprises | RHEL replacement, Production servers |
| **AlmaLinux** | RHEL | dnf, yum | Same as Rocky Linux | Secure enterprise workloads | RHEL-compatible open-source OS |
| **Red Hat Enterprise Linux (RHEL)** | RHEL | dnf, yum, rpm | dnf install, systemctl | Licensed servers in large companies, government | Mission-critical enterprise systems |

| | | | | | |
|---|---|---|---|---|---|
| **Fedora** | RHEL | dnf, rpm | dnf install, journalctl | Dev environments, testing new Linux features | Latest tech preview for RHEL |
| **Arch Linux** | Independent | pacman | pacman -S, systemctl | Developers, custom environments | Rolling release, bleeding-edge features |
| **Slackware** | Independent | pkgtool, installpkg | Manual package installation | Old-school sysadmin environments | Simplicity, educational use |
| **Gentoo** | Independent | emerge, Portage | emerge package | Custom hardware optimization | Performance-tuned systems |
| **Ubuntu** | Debian | apt, dpkg | apt install, service | Cloud servers, DevOps, CI/CD pipelines (Jenkins, Docker) | Beginners, Cloud (AWS, GCP), DevOps |
| **Debian** | Debian | apt, dpkg | apt-get install, systemctl | Servers, stability-focused systems | Stability, Debian-based distros like Ubuntu |
| **Kali Linux** | Debian | apt | apt install, nmap, airmon-ng | Cybersecurity, Penetration testing | Ethical hacking, cybersecurity training |
| **Manjaro** | Arch | pacman, pamac | pamac install, pacman -Syu | User-friendly Arch for developers | Easy Arch alternative |

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**Key Differences in Commands (Examples)**

| **Task** | **Ubuntu/Debian (apt)** | **RHEL/CentOS/Rocky (yum/dnf)** | **Arch (pacman)** | **openSUSE (zypper)** |
|---|---|---|---|---|
| **Install Package** | apt install nginx | dnf install nginx | pacman -S nginx | zypper install nginx |
| **Start Service** | systemctl start nginx | systemctl start nginx | systemctl start nginx | systemctl start nginx |
| **Check Firewall Rules** | ufw status | firewall-cmd --list-all | (manual config) | firewalld or yast |

| | | | | |
|---|---|---|---|---|
| **Add User** | adduser sakshi | adduser sakshi | useradd sakshi | useradd sakshi |
| **Update System** | apt update && apt upgrade | dnf update | pacman -Syu | zypper update |

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**Which Linux Flavor is Used Where?**

| **Industry/Use Case** | **Common Linux Flavors Used** |
|---|---|
| **Web Hosting (LAMP/LEMP)** | Ubuntu, CentOS, RHEL |
| **Enterprise IT** | RHEL, Rocky, Alma, SLES |
| **Cloud Providers** | Ubuntu, Debian, CentOS |
| **Penetration Testing** | Kali Linux |
| **DevOps & Containers** | Ubuntu, Debian, Rocky, Fedora |
| **SAP Environments** | SLES, openSUSE |
| **IoT / Embedded** | Yocto, Alpine, Debian |
| **Rolling Release Devs** | Arch, Manjaro |
| **User-Friendly Desktops** | Linux Mint, Pop!_OS, Zorin |

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Here's a **Linux Command Cheatsheet** tailored for real-time use in IT companies across popular Linux distributions like Ubuntu, RHEL, CentOS, Rocky, Debian, and more.

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**Linux Command Cheatsheet for IT Professionals**

**Package Management**

| Task | Ubuntu/Debian (apt) | RHEL/CentOS/Rocky (yum/dnf) | Arch (pacman) | openSUSE (zypper) |
|---|---|---|---|---|
| **Update repo cache** | sudo apt update | sudo dnf check-update | sudo pacman -Sy | sudo zypper refresh |
| **Upgrade packages** | sudo apt upgrade | sudo dnf upgrade | sudo pacman -Su | sudo zypper update |
| **Install package** | sudo apt install nginx | sudo dnf install nginx | sudo pacman -S nginx | sudo zypper install nginx |
| **Remove package** | sudo apt remove nginx | sudo dnf remove nginx | sudo pacman -R nginx | sudo zypper remove nginx |
| **Search package** | apt search nginx | dnf search nginx | pacman -Ss nginx | zypper search nginx |

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**System Services**

| Task | Universal Systemd (systemctl) |
|---|---|
| **Start a service** | sudo systemctl start nginx |
| **Stop a service** | sudo systemctl stop nginx |
| **Enable at boot** | sudo systemctl enable nginx |
| **Disable from boot** | sudo systemctl disable nginx |
| **Check status** | systemctl status nginx |
| **Restart service** | sudo systemctl restart nginx |

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**User and Permission Management**

| Task | Command |
|---|---|
| **Add user** | sudo adduser sakshi |
| **Delete user** | sudo deluser sakshi |
| **Add user to group** | sudo usermod -aG sudo sakshi |
| **List users** | cat /etc/passwd |
| **Change file ownership** | sudo chown user:group filename |
| **Change permissions** | chmod 755 filename |
| **View file permissions** | ls -l filename |

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**File System & Directory**

| Task | Command |
|---|---|
| **List files** | ls -l or ls –a |
| **Change directory** | cd /path/to/directory |
| **Copy file** | cp source dest |
| **Move file** | mv oldname newname |
| **Delete file** | rm filename |
| **Delete directory** | rm -r foldername |
| **View file content** | cat, less, more, tail –f |
| **Find files** | find /path -name filename |
| **Disk usage** | df -h (disk), du -sh * (dir size) |

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**Networking**

| Task | Command |
|---|---|
| **Show IP address** | ip a or ifconfig |
| **Check connectivity** | ping google.com |
| **DNS lookup** | nslookup example.com or dig |
| **Port scan** | nmap IP (Install: apt install nmap) |
| **Show listening ports** | ss -tuln or netstat –tuln |

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**Firewall (UFW & Firewalld)**

| Task (UFW - Ubuntu) | Command |
|---|---|
| **Enable firewall** | sudo ufw enable |
| **Allow port** | sudo ufw allow 80/tcp |
| **Deny port** | sudo ufw deny 22 |
| **Check status** | sudo ufw status |

| Task (Firewalld - RHEL) | Command |
|---|---|
| **Start firewall** | sudo systemctl start firewalld |
| **Allow port** | sudo firewall-cmd --add-port=80/tcp –permanent |
| **Reload rules** | sudo firewall-cmd –reload |
| **Check open ports** | sudo firewall-cmd --list-all |

**Process & Resource Monitoring**

| Task | Command |
|---|---|
| **View running processes** | ps aux, top, htop |
| **Kill process** | kill PID, killall nginx |
| **CPU/Memory usage** | top, htop |
| **Check uptime** | Uptime |
| **Monitor real-time logs** | tail -f /var/log/syslog |

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**Others (Handy)**

| Task | Command |
|---|---|
| **Switch to root** | sudo -i or su – |
| **Show kernel version** | uname –r |
| **Show OS version** | cat /etc/os-release |
| **Reboot system** | sudo reboot |
| **Shutdown system** | sudo shutdown now |

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Here's a **Linux Command Cheat Sheet** organized by common tasks, comparing the commands across major Linux flavors:

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**Linux Command Cheat Sheet by Distro**

**1. Package Management**

| Task | Ubuntu/Debian (apt) | RHEL/CentOS/Rocky (dnf/yum) | Arch (pacman) | openSUSE (zypper) |
|---|---|---|---|---|
| **Update package list** | sudo apt update | sudo dnf check-update | sudo pacman -Sy | sudo zypper refresh |
| **Upgrade packages** | sudo apt upgrade | sudo dnf upgrade | sudo pacman -Syu | sudo zypper update |
| **Install a package** | sudo apt install nginx | sudo dnf install nginx | sudo pacman -S nginx | sudo zypper install nginx |
| **Remove a package** | sudo apt remove nginx | sudo dnf remove nginx | sudo pacman -R nginx | sudo zypper remove nginx |
| **Clean cache** | sudo apt clean | sudo dnf clean all | sudo pacman -Sc | sudo zypper clean |

**2. Service Management (Systemd)**

| Task | Command |
|---|---|
| **Start a service** | sudo systemctl start nginx |
| **Stop a service** | sudo systemctl stop nginx |
| **Restart a service** | sudo systemctl restart nginx |
| **Check status** | sudo systemctl status nginx |
| **Enable on boot** | sudo systemctl enable nginx |
| **Disable on boot** | sudo systemctl disable nginx |

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### 3. User Management

| Task | Command |
|---|---|
| **Add a user** | sudo adduser sakshi (Debian/Ubuntu) sudo useradd sakshi (RHEL/Arch) |
| **Delete a user** | sudo deluser sakshi (Debian/Ubuntu) sudo userdel sakshi (RHEL/Arch) |
| **Add user to group** | sudo usermod -aG docker sakshi |
| **Change user password** | sudo passwd sakshi |

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### 4. File & Directory

| Task | Command |
|---|---|
| **List files** | ls, ls -l, ls –la |
| **Change directory** | cd /path/to/dir |
| **Make directory** | mkdir new_folder |
| **Copy files** | cp source.txt destination.txt |
| **Move/Rename files** | mv old.txt new.txt |
| **Remove file** | rm file.txt |
| **Remove directory** | rm -r folder |

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### 5. Network Commands

| Task | Command |
|---|---|
| **Check IP address** | ip a or ifconfig (needs net-tools) |
| **Check routing table** | ip route |
| **Ping a host** | ping google.com |
| **Download via CLI** | wget <url> or curl -O <url> |

| | |
|---|---|
| **Open firewall port** | sudo ufw allow 80 (Ubuntu) sudo firewall-cmd --add-port=80/tcp --permanent && sudo firewall-cmd --reload (RHEL) |

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### 6. Process & Resource Monitoring

| Task | Command |
|---|---|
| **Show running processes** | ps aux |
| **Monitor system usage** | top, htop (install required) |
| **Check memory usage** | free –h |
| **Disk usage** | df –h |
| **Directory size** | du -sh /path |
| **Kill process by PID** | kill -9 <pid> |

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### 7. System Management

| Task | Command |
|---|---|
| **Reboot system** | sudo reboot |
| **Shutdown system** | sudo poweroff |
| **Check system uptime** | Uptime |
| **System info** | uname -a, lsb_release -a (Ubuntu) |

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### 8. Archive & Compression

| Task | Command |
|---|---|
| **Create a .tar.gz** | tar -czvf archive.tar.gz folder/ |
| **Extract a .tar.gz** | tar -xzvf archive.tar.gz |
| **Zip a folder** | zip -r folder.zip folder/ |

| | |
|---|---|
| **Unzip a file** | unzip file.zip |

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**9. Search & Find**

| Task | Command |
|---|---|
| **Search for a file** | find / -name filename.txt |
| **Search inside files** | grep "text" file.txt |
| **Recursive grep** | grep -r "text" /path/ |
| **Count lines/words** | wc -l file.txt, wc -w file.txt |

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Great! Below is the **Linux command cheatsheet in .bashrc alias and man-page-style**, perfect for terminal usage, scripting, and power users.

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**1. Useful .bashrc Aliases for Real-Time IT Work**

Add these lines to your ~/.bashrc or ~/.bash_aliases, then run source ~/.bashrc.

```
# --- Package Management ---
alias update="sudo apt update && sudo apt upgrade -y"
alias install="sudo apt install"
alias remove="sudo apt remove"
alias search="apt search"
alias cleanup="sudo apt autoremove && sudo apt autoclean"

# RHEL/CentOS/DNF-based
alias yupdate="sudo dnf update -y"
alias yinstall="sudo dnf install"
alias yremove="sudo dnf remove"

# Arch
alias pacupd="sudo pacman -Syu"
alias pacin="sudo pacman -S"

# openSUSE
alias zupdate="sudo zypper update"
alias zinstall="sudo zypper install"

# --- System Services ---
alias start="sudo systemctl start"
```

```
alias stop="sudo systemctl stop"
alias restart="sudo systemctl restart"
alias status="systemctl status"
alias enable="sudo systemctl enable"
alias disable="sudo systemctl disable"

# --- Networking ---
alias ports="ss -tuln"
alias myip="ip a"
alias pingg="ping google.com"
alias ns="nslookup"

# --- Users and Permissions ---
alias users="cat /etc/passwd"
alias groups="cat /etc/group"
alias chperm="chmod 755"
alias chownme="sudo chown $USER:$USER"

# --- Logs ---
alias syslog="sudo tail -f /var/log/syslog"
alias dmesglog="dmesg | less"
alias journal="journalctl -xe"

# --- Disk and File ---
alias usage="df -h"
alias dusize="du -sh *"
alias findf="find . -name"

# --- Git Shortcuts ---
alias gs="git status"
alias ga="git add ."
alias gc="git commit -m"
alias gp="git push"
alias gl="git log --oneline --graph"

# --- Docker Shortcuts ---
alias dps="docker ps"
alias dim="docker images"
alias dstart="docker start"
alias dstop="docker stop"
alias drm="docker rm"
alias drmi="docker rmi"

# --- Kubernetes Shortcuts ---
alias k="kubectl"
alias kgp="kubectl get pods"
alias kgs="kubectl get svc"
alias kgn="kubectl get nodes"
```

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**2. Cheatsheet in Man Page Format**

# Example Usage: man cheatsheet

To create your own man-style page:

1. Create the file: sudo nano /usr/local/man/man1/cheatsheet.1
2. Paste this content:

```
.TH CHEATSHEET 1 "August 2025" "Linux Admin Guide" "User Commands"

.SH NAME
cheatsheet \- Linux admin command reference for daily IT operations

.SH SYNOPSIS
cheatsheet

.SH DESCRIPTION
A short reference guide with frequently used commands across different Linux distributions
(Ubuntu, RHEL, Arch, openSUSE).

.SH COMMANDS

.TP
.B Package Management (Ubuntu/Debian)
apt update && apt upgrade -y
apt install <package>
apt remove <package>

.TP
.B Package Management (RHEL/CentOS)
dnf update -y
dnf install <package>
dnf remove <package>

.TP
.B Service Management
systemctl start|stop|restart <service>
systemctl enable|disable <service>

.TP
.B Networking
ip a
ping google.com
ss -tuln
firewall-cmd --list-all

.TP
```

```
.B File and User
chmod 755 <file>
chown user:group <file>
adduser <username>

.TP
.B Logs
tail -f /var/log/syslog
journalctl -xe

.SH AUTHOR
Sakshi Virha (sakshivirhajobsearch@yahoo.com)

.SH NOTES
For more commands, refer to 'man <command>'.
```

3. Compress and index:

```
gzip /usr/local/man/man1/cheatsheet.1
mandb
```

4. Use it:

```
man cheatsheet
```

---

Would you like a **PDF version** of this .bashrc + man-page cheatsheet or a **ready-to-import bashrc file**?